

Crane Lifting Plan

Load chart, daily crane check, lift plan and tandem lift in one screen

⚠ FOR UNDERSTANDING PURPOSES ONLY

Prasanaa Kartick T — Personal Learning Project

Flutter / Dart

Load chart

Daily check

Lift plan

Tandem lift

1. What It Is

A mobile crane lift is only safe if the load, the crane and the ground all agree. This screen puts everything needed to plan one lift into **four tabs**, and each can be saved and exported as a PDF. It is reached from Safety › Process Safety › Lifting Operations › Crane Lifting Plan.

Tab	Purpose
Load Chart	Describe the crane and enter its rated capacities
Daily Check	The crane operator's daily inspection checklist, with a PDF
Lift Plan	Calculate whether a single-crane lift is safe, and record the authorisations
Tandem	Plan a lift shared between two cranes

2. Load Chart Tab

The load chart is the crane maker's table of how much the crane may lift at each boom length and working radius. The rest of the screen depends on it.

- **Crane details:** type (crawler or mobile), make, model, capacity (default 50 tonnes) and licence plate.
- **The chart:** a grid of boom length (metres) against radius (metres) holding the rated capacity. A **default load chart** is supplied. Press **Edit** to change the values, and **Done** to lock them. A **Reset** button restores the default.
- **Save and open.** Several load charts can be saved under names, for example one per crane, and opened later. If a saved chart is damaged, a message says "Open failed: invalid saved data" instead of crashing.

The chart feeds everything else

The Lift Plan and Tandem tabs read the crane's rated capacity from this chart. If the chart has no value for the chosen boom length and radius, the plan shows **NO LIFT** instead of guessing.

3. Daily Check Tab

A **Daily Mobile Crane Inspection Checklist** in eight numbered groups, about ninety-six items in all. Each item cycles through **N/A**, **OK** and **Not OK** with a tap and has a remarks box.

Group	Examples
1. Pre-start walk-around	Inspection sticker, doors and panels, spark arrester, leaks under the chassis, tyres or tracks, hoses, coolant and engine oil, wire rope, sheaves and hook block, jib, load chart accessible, fire extinguisher
2. Engine start-up	Engine start, voltmeter, tachometer, beacon and signal lights, horn and wipers, oil pressure, coolant temperature, fuel gauge
3. Crane safety checks	Brakes, transmission oil, air and hydraulic filters, suspension, anti-two-block , boom angle and load limiter , outriggers and float pads, bubble level, steering, swing lock
4. Cabin items	Operator chair, cabin seals and door, wiper, lights, fan, boom angle indicator
5. Control and safety systems	Load indicator, pressure gauges, hour meters, emergency shutdown valve, main switch
6. Electrical and mechanical	Lights, cables, nuts and bolts, wedges and keys, hook-block latches, brake linings
7. Pneumatic and hydraulic	Hoses, pipes, fittings, accumulators, drain valves, filters, main hydraulic pump
8. Wire ropes	Main hoist, auxiliary hoist and luff winch ropes: kinks, breaks, wear, drum lay and terminations

The PDF has two parts: **Section A, the inspection**, and **Section B, the endorsement**, where the inspector signs, corrective actions are recorded and an approver signs. The crane details appear at the top of the first page.

4. Lift Plan Tab

Begins with a **setup checklist** to be completed by the crane operator, then the numbers.

Part	What you enter or see
Boom geometry	Boom length (A), working radius (B), hook height (C). A diagram of the crane and boom is drawn and the boom angle is worked out

Part	What you enter or see
Load inputs	Hook-block weight, material weight and total weight (D), in tonnes or kilograms
Outrigger parameters	Outrigger distance, pad area and soil capacity
Rated capacity	Read from the load chart for that boom length and radius
Load moment and outrigger load	Calculated from the load and the geometry
Crane % capacity	Total load ÷ rated capacity × 100
Ground bearing pressure	The load carried by the outrigger pads, compared with the soil capacity

Reading the result

Crane % capacity	Lifting status
75% or less	Safe (green)
Above 75% and below 90%	Caution (amber), showing the percentage
90% or more	Critical (red)
No load-chart value	NO LIFT

Ground status reads Safe when the ground pressure is below the soil capacity entered. The plan finishes with authorisation lines for the lifting supervisor, crane operator, competent rigger and HSE in-charge, each with name, ID, time and date, and exports as a one-page PDF.

A lift plan can be saved by name and opened again from a list of saved plans.

5. Tandem Tab

A **tandem lift** shares one load between two cranes. The tab lets you pick a saved load chart for **Crane A** and **Crane B** and enter the lift:

- **Net load** (the object's weight), the **total span** between the two hooks, the **centre-of-gravity distance** from Crane A and the height of the lifted object.
- For each crane: radius, boom length and angle, hook-block weight, the **share** of the load, and its **gross load**.
- A **tandem diagram** and a boom-position panel are drawn from the entries.

The sharing rule

Each crane's share is worked out from the distance between the centre of gravity (G) and the **opposite** crane: load on A = total weight $\times$ (distance from G to B) $\div$ total span, and load on B = total weight $\times$ (distance from G to A) $\div$ total span. The crane closer to G therefore carries the larger share. The tab then shows each crane's percentage of capacity and raises an alarm: **both cranes must stay below 90% of capacity**. The check list must also be marked complete.

Worked examples

A 30-tonne load with the cranes 12 metres apart:

Position of G	Load on Crane A	Load on Crane B
In the middle (6 m from each crane)	$30 \times 6 \div 12 = 15 \text{ t}$	$30 \times 6 \div 12 = 15 \text{ t}$
4 m from A and 8 m from B	$30 \times 8 \div 12 = 20 \text{ t}$	$30 \times 4 \div 12 = 10 \text{ t}$
3 m from A and 9 m from B	$30 \times 9 \div 12 = 22.5 \text{ t}$	$30 \times 3 \div 12 = 7.5 \text{ t}$
Directly under Crane A	$30 \times 12 \div 12 = 30 \text{ t}$	0 t

The two shares always add up to the full load. The same rule applies to a tailing lift, where a main crane lifts and a tailing crane controls the tail: the crane nearer the centre of gravity carries more.

The PDF includes crane A and crane B details, the shared load, and authorisation lines for each crane operator, the competent rigger and the HSE in-charge.

6. Saving

Load charts, lift plans and tandem lifts each have their own saved lists in the app database, so a plan prepared today can be opened and adjusted tomorrow. The permit composite export can also pull a saved lift plan or tandem plan into its bundle.